Lecture 6 Model Selection II

STAT 441: Statistical Methods for Learning and Data Mining

Wenlu TANG

Department of Mathematical and Statistical Sciences
University of Alberta

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Outline

1. Ridge Regression
2. The LASSO
3. Ridge regression and the LASSO
4. Summary and Remark

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Shrinkage Methods

Ridge regression and Lasso

- The subset selection methods use least squares to fit a linear model that contains a subset of the predictors.
- As an alternative, we can fit a model containing all p predictors using a technique that constrains or regularizes the coefficient estimates, or equivalently, that shrinks the coefficient estimates towards zero.
- It may not be immediately obvious why such a constraint should improve the fit, but it turns out that shrinking the coefficient estimates can significantly reduce their variance.

Least squares is sometimes sensitive

Model

$$Y = Z\beta + \varepsilon$$

Suppose $Z'Z$ invertible. Least squares estimate:

$$\hat{\beta} = (Z'Z)^{-1}Z'y$$

$$Z = \underbrace{\begin{bmatrix} Z_{1,1} & \cdots & Z_{1,r+1} \\ \vdots & \ddots & \vdots \\ Z_{n,1} & \cdots & Z_{n,r+1} \end{bmatrix}}_{n\times(r+1)} = U\Sigma V' \quad \text{SVD}$$

Non sq matrix

$$= \underbrace{\begin{bmatrix} U_{1,1} & \cdots & U_{1,n} \\ \vdots & \ddots & \vdots \\ U_{n,1} & \cdots & U_{n,n} \end{bmatrix}}_{U,\ n\times n} \underbrace{\begin{bmatrix} \sigma_1 & & 0 \\ & \ddots & \\ 0 & & \sigma_{r+1} \\ 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{bmatrix}}_{\Sigma,\ n\times(r+1)} \underbrace{\begin{bmatrix} V_{1,1} & \cdots & V_{r+1,1} \\ \vdots & \ddots & \vdots \\ V_{1,r+1} & \cdots & V_{r+1,r+1} \end{bmatrix}}_{V',\ (r+1)\times(r+1)}.$$

Pseudo diagonal matrix

Sigma - eigenvalue of Z^tZ

Eigenvalue decomposition

$$Z'Z = V\Sigma'U'U\Sigma V' = V\Sigma'\Sigma V' = \overbrace{V\mathrm{diag}\{\sigma_1^2, \cdots, \sigma_{r+1}^2\}V'}$$

$$= \underbrace{V \begin{bmatrix} \sigma_1 & & & 0 & \cdots & 0 \\ & \ddots & & \vdots & \ddots & \vdots \\ & & \sigma_{r+1} & 0 & \cdots & 0 \end{bmatrix}}_{\Sigma',\ (r+1)\times n} \underbrace{\begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_{r+1} \\ 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{bmatrix}}_{\Sigma,\ n\times(r+1)} V'$$

$$(Z'Z)^{-1} = V\mathrm{diag}\left\{\frac{1}{\sigma_1^2}, \frac{1}{\sigma_2^2}, \cdots, \frac{1}{\sigma_{r+1}^2}\right\} V' = V(\Sigma'\Sigma)^{-1}V'$$

$$(Z'Z)^{-1}Z' = V\mathrm{diag}\left\{\frac{1}{\sigma_1^2}, \frac{1}{\sigma_2^2}, \cdots, \frac{1}{\sigma_{r+1}^2}\right\} \underbrace{\begin{bmatrix} \sigma_1 & & & 0 & \cdots & 0 \\ & \ddots & & \vdots & \ddots & \vdots \\ & & \sigma_{r+1} & 0 & \cdots & 0 \end{bmatrix}}_{\Sigma',\ (r+1)\times n} U'$$

$$
\begin{aligned}
(Z'Z)^{-1}Z'y =& V \underbrace{\begin{bmatrix} \sigma_1^{-1} & & & 0 & \cdots & 0 \\ & \ddots & & \vdots & \ddots & \vdots \\ & & \sigma_{r+1}^{-1} & 0 & \cdots & 0 \end{bmatrix}}_{(\Sigma'\Sigma)^{-1}\Sigma',\ (r+1)\times n} \underbrace{\begin{bmatrix} (U'y)_1 \\ \vdots \\ (U'y)_n \end{bmatrix}}_{U'y,\ n\times 1} \\
=& V \underbrace{\begin{bmatrix} \sigma_1^{-1}(U'y)_1 \\ \vdots \\ \sigma_{r+1}^{-1}(U'y)_{r+1} \end{bmatrix}}_{(\Sigma'\Sigma)^{-1}\Sigma'U'y,\ n\times 1}
\end{aligned}
$$

Note that some singular values σ_i could be very small, especially when, e.g., two columns of Z are very close (collinear).

Ridge regression can solve collinearity

Solution, which is also one of the motivations of the ridge regression: change $(Z'Z)^{-1}$ to

$$(Z'Z + \lambda I)^{-1} = V \quad \text{diag}\left\{\frac{1}{\sigma_1^2 + \lambda}, \cdots, \frac{1}{\sigma_{r+1}^2 + \lambda}\right\} \quad V'$$

Add some constant to avoid it going to infinfity

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Ridge Regression

- The ridge regression coefficient estimates $\hat{\beta}^R$ are the values that minimize

L2 penalty

$$\sum_i \left(y_i - \beta_0 - \sum_j \beta_j x_{ij} \right)^2 + \lambda \sum_j \beta_j^2,$$

If lambda goes to inf beta goes to zero

where λ is a tuning parameter, to be determined separately.

- The second term $\lambda \sum_j \beta_j^2$ called a shrinkage penalty, is small when β_j, $j \geq 1$ are close to zero, and so it has the effect t of shrinking the estimates of β_j towards zero.
- The tuning parameter λ serves to control the relative impact of these two terms on the regression coefficient estimates.
- Selecting a good value for λ is critical; cross-validation is used for this.

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About the model without intercept

To solve the ridge regression problem, we let

$$\mathcal{E}(\beta) = \frac{1}{n}\|y - Z\beta\|^2 + \lambda\|\beta\|^2,$$

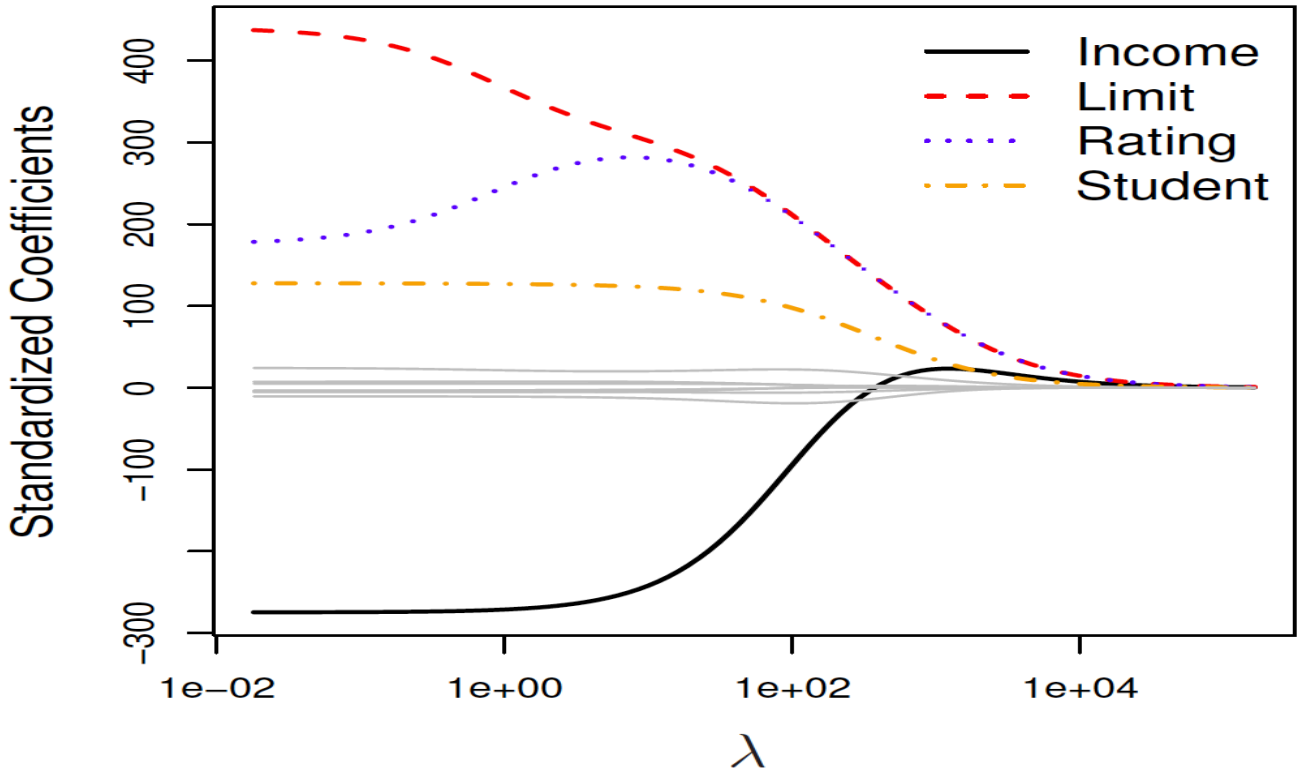
then we take gradient.

$$D_\beta \mathcal{E}(\beta) = \frac{2}{n} Z'(Z\beta - y) + 2\lambda\beta.$$

Let the gradient equal to zero to give

$$\hat{\beta}^{\text{ridge}} = (Z'Z + n\lambda I)^{-1} Z'y.$$

Credit data example



In the left-hand panel, each curve corresponds to the ridge regression coefficient estimate for one of the ten variables, plotted as a function of λ. As λ increases, the coefficients are shrunken to zeros.

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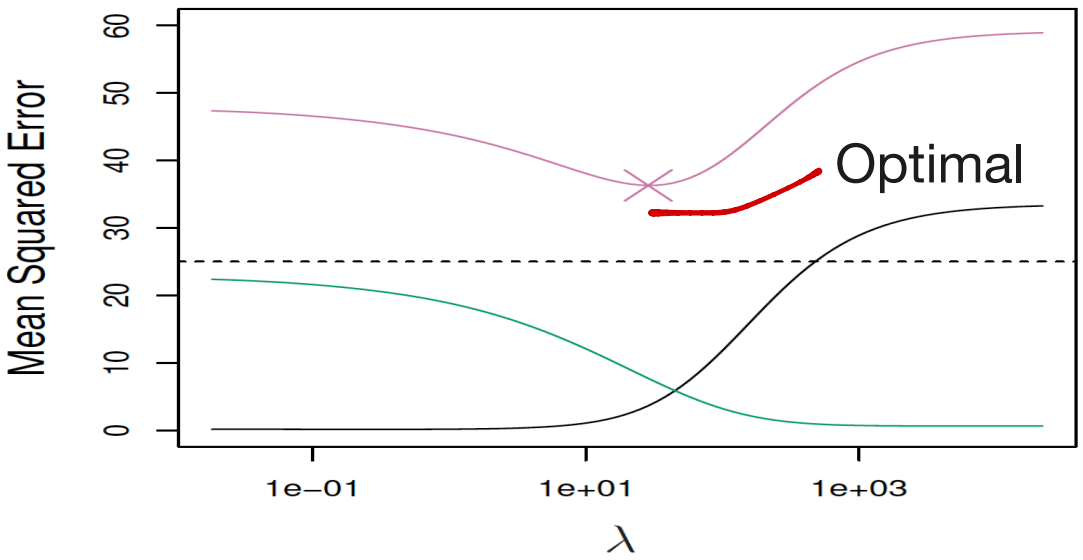
Scaling of predictors

- The standard least squares coefficient estimates are scale equivariant: multiplying X_j by a constant c simply leads to a scaling of the least squares coefficient estimates by a factor of $1/c$. In other words, regardless of how the j-th predictor is scaled $X_j\hat{\beta}_j$ will remain the same.
- In contrast, the ridge regression coefficient estimates can change substantially when multiplying a given predictor by a constant, due to the sum of squared coefficient term in the penalty part of the ridge regression objective function.
- Therefore, it is best to apply ridge regression after standardizing the predictors, using the formula

$$\tilde{x}_{ij} = x_{ij} \Big/ \sqrt{\sum_i (x_{ij} - \bar{x}_j)^2 / n}.$$

Credit data example



Simulated data with $n = 50$ observations, $p = 45$ predictors, all having nonzero coefficient. Squared bias (black), variance (green), and test mean squared error (purple) for the ridge regression predictions on a simulated data set. The horizontal dashed lines indicate the minimum possible MSE. The purple crosses indicate the ridge regression models for which the MSE is smallest.

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The LASSO

- Ridge regression does have one obvious disadvantage: unlike subset selection, which will generally select models that involve just a subset of the variables, ridge regression will include all p predictors in the final model.
- The LASSO is a relatively recent alternative to ridge regression that overcomes this disadvantage. The lasso coefficient $\hat{\beta}^L$ minimize the quantity

$$\sum_i \left(y_i - \beta_0 - \sum_j \beta_j x_{ij} \right)^2 + \lambda \sum_j |\beta_j|,$$

where λ is a tuning parameter.

Overcomes the disadvantage to of electing all predictors in ridge

- The LASSO uses l_1 penalty instead of l_2 (ridge regression).
- ℓ_1 norm of a coefficient vector β: $\|\beta\|_1 = \sum_{j=1}^p |\beta_j|$; ℓ_2 norm of a coefficient vector β: $\|\beta\|_2 = \sqrt{\sum_{j=1}^p \beta_j^2}$.

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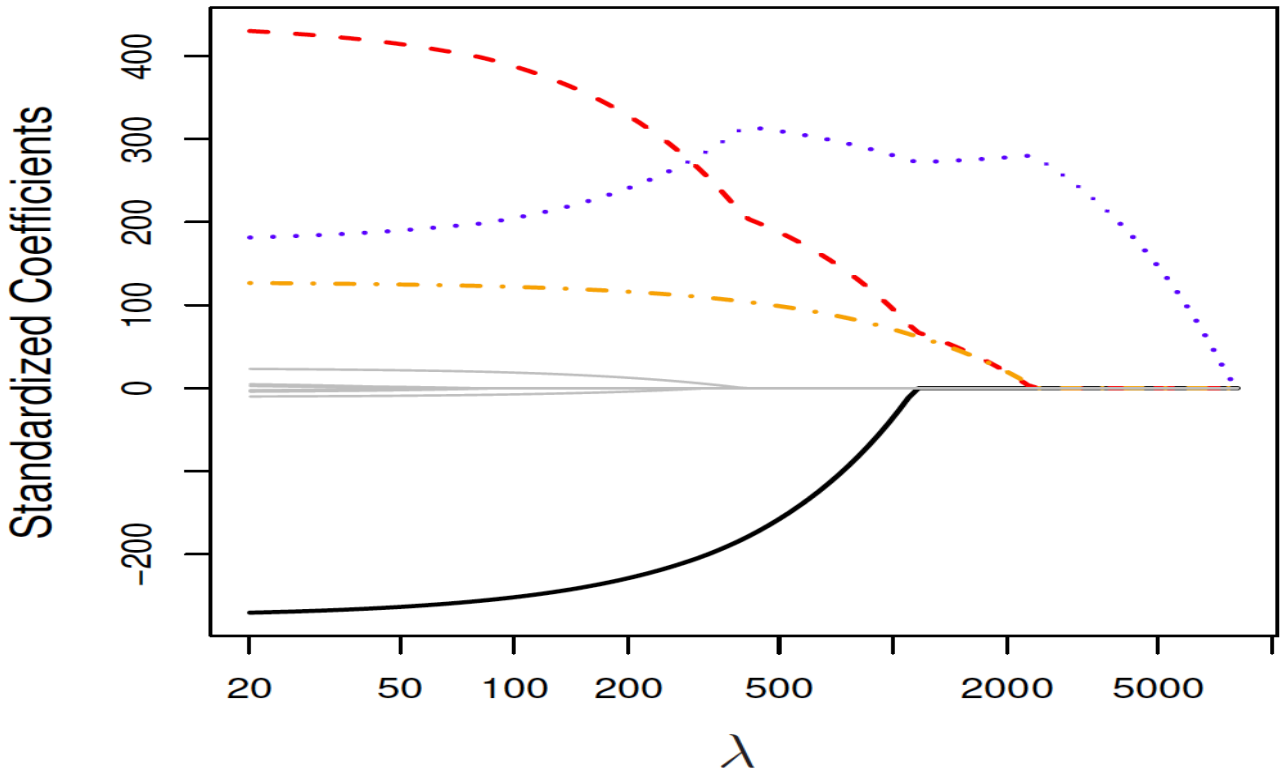
The LASSO

- As with ridge regression, the lasso shrinks the coefficient estimates towards zero as λ increases.
- However, in the case of the lasso, the l_1 penalty has the effect of forcing some of the coefficient estimates to be exactly equal to zero when the tuning parameter λ is sufficientl large. Thus performs variable selection.
- We say that the lasso yields sparse models — that is, models that involve only a subset of the variables.
- Selecting a good value for λ is critical; cross-validation is again used for this.

Credit data example



As λ increases, the coefficients are shrunken to exact zeros.

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LASSO Solution: soft thresholding

For the orthogonal design case with $X'X = I$, the LASSO method is equivalent to: solve β_j 's componentwisely by solving

$$\min_{\beta_j} \left(\beta_j - \hat{\beta}_j^{ls}\right)^2 + \lambda|\beta_j|.$$

Lasso - strongly push the small values to 0

The solution to the above problem is

$$\hat{\beta}_j^{\text{lasso}} = \text{sign}\left(\hat{\beta}_j^{ls}\right)\left(\left|\hat{\beta}_j^{ls}\right| - \frac{\lambda}{2}\right)_+$$

(X)+ = max(0,X)

Keep sign , remove value

$$= \begin{cases} \hat{\beta}_j^{ls} - \frac{\lambda}{2} & \text{if} \quad \hat{\beta}_j^{ls} > \frac{\lambda}{2} \\ 0 & \text{if} \quad \left|\hat{\beta}_j^{ls}\right| \leq \frac{\lambda}{2} \\ \hat{\beta}_j^{ls} + \frac{\lambda}{2} & \text{if} \quad \hat{\beta}_j^{ls} < -\frac{\lambda}{2} \end{cases}$$

- shrinks big coefficients by a constant $\frac{\lambda}{2}$ towards zero.
- truncates small coefficients to zero exactly

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Comparison: Different Methods Under Orthogonal Design

When X is orthonormal, i.e. $X^T X = I$

- Best subset (of size M) : $\hat{\beta}_j^{ls}$.
  - keeps the largest coefficients.
- Ridge regression: $\hat{\beta}_j^{ls}/(1+\lambda)$
  - does a proportional shrinkage
- Lasso: $\text{sign}\left(\hat{\beta}_j^{ls}\right)\left(\hat{\beta}_j^{ls} - \frac{\lambda}{2}\right)_+$
  - transform each coefficient by a constant factor first, then truncate it at zero with a certain threshold.
  - "soft thresholding", used often in wavelet-based smoothing.

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Ridge regression and the LASSO

- Why is it that the lasso, unlike ridge regression, results in coefficient estimates that are exactly equal to zero?
- One can show that the lasso and ridge regression coefficient estimates solve the problems

$$\min_{\beta} \sum_{i} \left(y_i - \beta_0 - \sum_{j} \beta_i x_{ij} \right)^2, \text{ subject to } \sum_{j} |\beta_j| \leq c;$$

$$\min_{\beta} \sum_{i} \left(y_i - \beta_0 - \sum_{j} \beta_i x_{ij} \right)^2, \text{ subject to } \sum_{j} \beta_j^2 \leq c;$$

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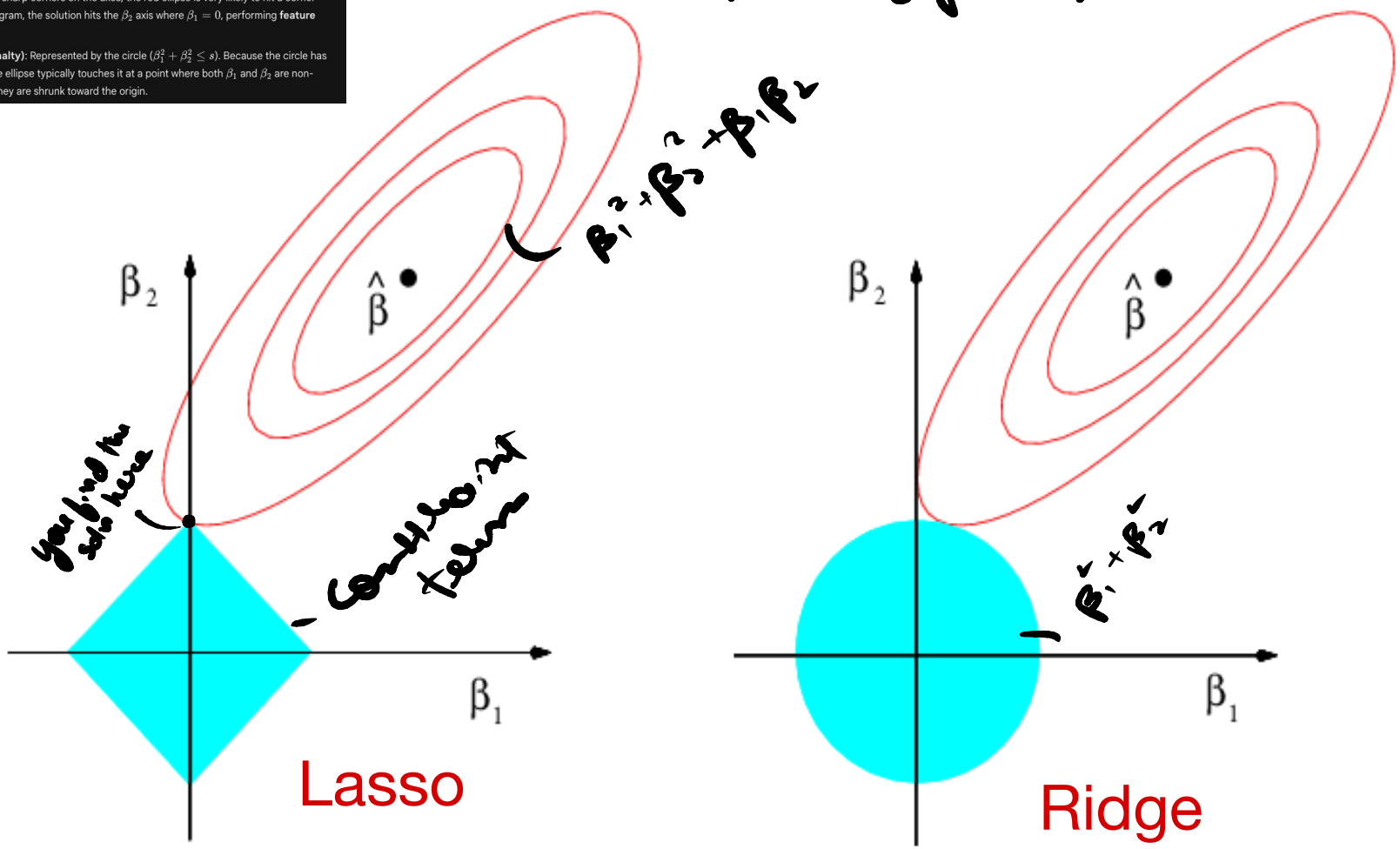
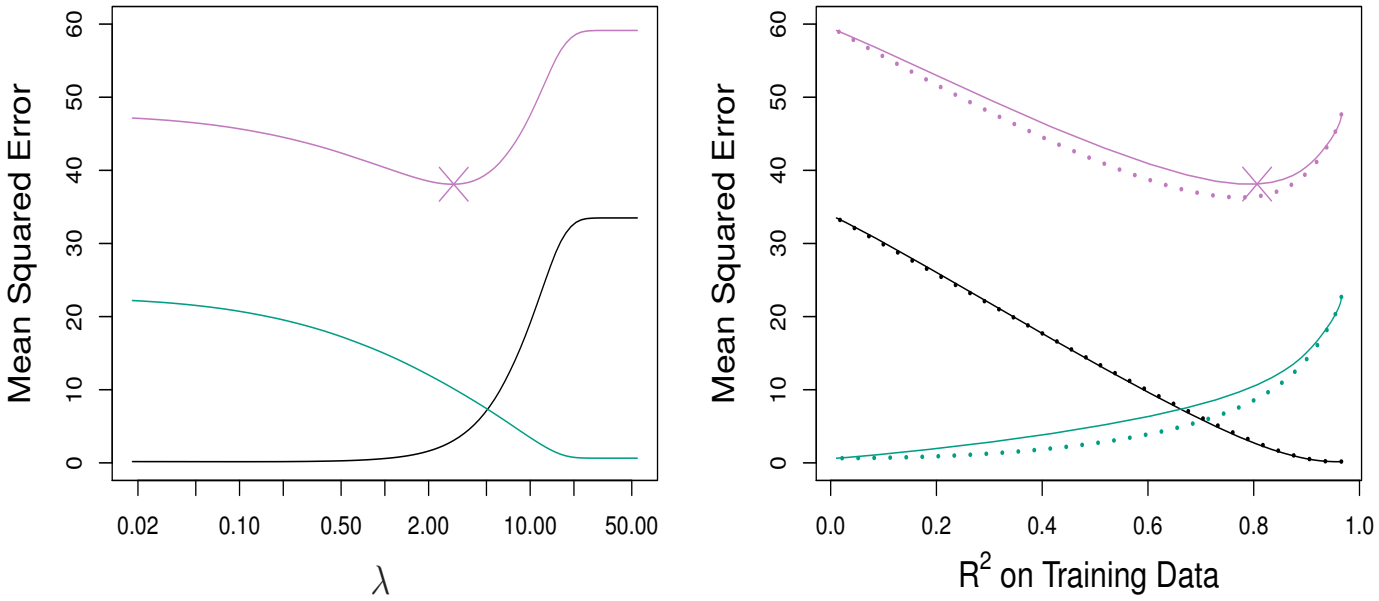
Ridge regression and the LASSO

- **Constraint Regions (Blue Shapes):**
  - **LASSO (L_1 penalty):** Represented by the diamond shape ($|\beta_1| + |\beta_2| \leq s$). Because this shape has sharp corners on the axes, the red ellipse is very likely to hit a corner first. In the diagram, the solution hits the β_2 axis where $\beta_1 = 0$, performing **feature selection**.
  - **Ridge (L_2 penalty):** Represented by the circle ($\beta_1^2 + \beta_2^2 \leq s$). Because the circle has no corners, the ellipse typically touches it at a point where both β_1 and β_2 are non-zero, though they are shrunk toward the origin.

elipse – like an objective function

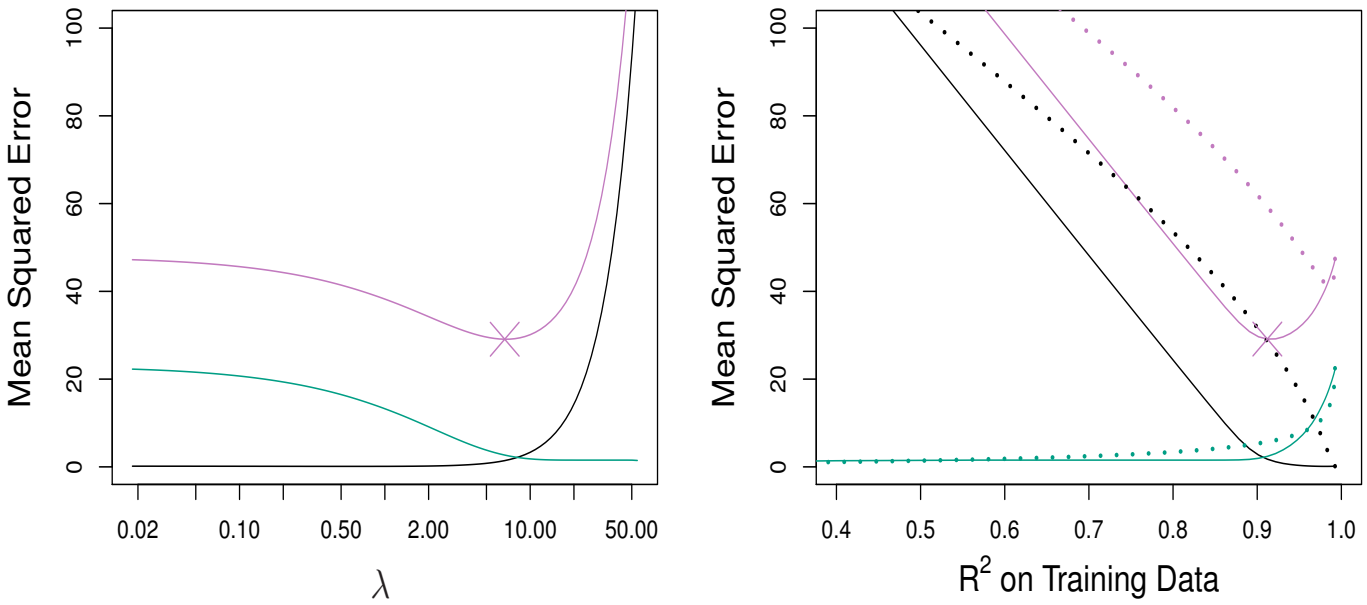


Credit data example



Left: Plots of squared bias (black), variance (green), and test mean squared error (purple) for the LASSO on a simulated data set. Right: Comparison of squared bias, variance and test MSE between lasso (solid) and ridge (dashed). The purple crosses indicate the LASSO models for which the MSE is smallest.

Credit data example



Left: Plots of squared bias (black), variance (green), and test mean squared error (purple) for the LASSO on another simulated data set. Right: Comparison of squared bias, variance and test MSE between lasso (solid) and ridge (dashed). The purple crosses indicate the LASSO models for which the MSE is smallest.

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Conclusions

- These two examples illustrate that neither ridge regression nor the lasso will universally dominate the other.
- In general, one might expect the lasso to perform better when the response is a function of only a relatively small number of predictors.
- However, the number of predictors that is related to the response is never known a priori for real data sets.
- A technique such as cross-validation can be used in order to determine which approach is better on a particular data set.

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Selecting the Tuning Parameter for Ridge Regression and Lasso

- As for subset selection, for ridge regression and lasso we require a method to determine which of the models under consideration is best.
- That is, we require a method selecting a value for the tuning parameter λ or equivalently, the value of the constraint c.
- Cross-validation provides a simple way to tackle this problem. We choose a grid of λ values, and compute the cross-validation error rate for each value of λ.
- We then select the tuning parameter value for which the cross-validation error is smallest.
- Finally, the model is re-fit using all of the available observations and the selected value of the tuning parameter.

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Summary and Remark

- Ridge Regression
- The LASSO
- Ridge Regression and the LASSO
- Read textbook Chapter 6